

Comprehensive Notes

Chapter 3: Diode Models and Circuits

In-Depth Explanations with Analogies, Graphs & Physical Intuition

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1 The Big Picture: Why Diodes?

The Cellphone Charger Problem (Fig. 3.1)

Problem: Your wall outlet provides 110 V AC at 60 Hz. Your phone needs 3.5 V DC. How?

Step 1: A **transformer** steps 110 V down to ~ 4 V (still AC).

Step 2: A mysterious “black box” must convert AC $\rightarrow$ DC.

The key insight: AC has equal positive and negative areas $\Rightarrow$ average = 0. If we could *remove* the negative half-cycles, the average becomes positive $\Rightarrow$ we get DC!

But a resistor can't do this. Ohm's law ($V = IR$) is linear: if input flips sign, output flips too. We need a **nonlinear** device — the **diode**.

The Turnstile Analogy

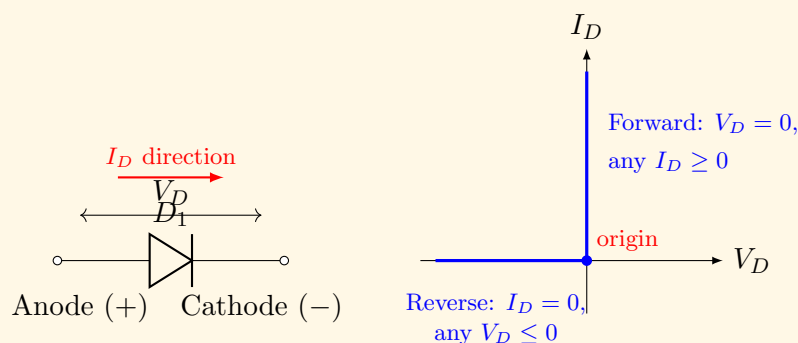
Think of a subway turnstile: it rotates freely when you push from the front (forward bias) but locks when pushed from behind (reverse bias). A diode does exactly this for electrical current — lets it through in one direction, blocks it in the other.

2 Section 3.1: The Ideal Diode

2.1 What is an Ideal Diode?

An ideal diode is a two-terminal device that acts as a **perfect one-way switch**: zero resistance in one direction, infinite resistance in the other.

Ideal Diode Symbol, I/V Characteristic & Water Analogy



The Two States — Memorize This!

State	Condition	Equivalent	What it means
ON (Forward)	$V_{anode} > V_{cathode}$	Short circuit	Wire with $V_D = 0$
OFF (Reverse)	$V_{anode} < V_{cathode}$	Open circuit	Gap with $I_D = 0$

Water Pipe Valve (Fig. 3.3c from textbook)

Imagine a pipe with a hinged flap valve and a stopper:

- **Water from left (forward):** Pushes flap open $\rightarrow$ water flows freely. *This is like $V_{anode} > V_{cathode}$.*
- **Water from right (reverse):** Pushes flap against stopper $\rightarrow$ sealed shut. *This is like $V_{anode} < V_{cathode}$.*

The flap has zero mass (ideal) $\rightarrow$ opens with any positive pressure (even tiny). In reality, a real diode needs $\sim 0.7\text{ V}$ “pressure” to open (constant-voltage model).

2.2 Understanding the I/V Characteristic

Why does the I/V look like an “L” shape?

The L-shape combines two limiting cases of Ohm’s law:

- $R = 0$ (**short circuit**): $I = V/0 = \infty$ for any V . This gives the **vertical line** at $V_D = 0$.
- $R = \infty$ (**open circuit**): $I = V/\infty = 0$ for any V . This gives the **horizontal line** at $I_D = 0$.

The ideal diode uses the vertical part for $V_D \geq 0$ and the horizontal part for $V_D \leq 0$.

Does $I_D = \infty$ really happen? No! In real circuits, the surrounding components (resistors, sources) limit the current. The diode simply sustains $V_D = 0$, and the external circuit determines I_D . Think of it like a wire: a wire can carry any current.

2.3 The Assume-and-Verify Method

CRITICAL TECHNIQUE — Used in EVERY Diode Problem

This is the single most important method in this chapter. Every exam problem uses it.

1. **ASSUME** the state of each diode (ON or OFF).
2. **REPLACE**:
 - ON diodes $\rightarrow$ **short circuit** (ideal) or $V_{D,on}$ **battery** (CV model)
 - OFF diodes $\rightarrow$ **open circuit** (remove from circuit)
3. **SOLVE** the resulting linear circuit using KVL/KCL (no diodes left!).
4. **VERIFY** your assumption:
 - Each ON diode: Is $I_D > 0$ flowing anode $\rightarrow$ cathode? If **NO** $\rightarrow$ contradiction!
 - Each OFF diode: Is $V_D < 0$ (ideal) or $V_D < V_{D,on}$ (CV)? If **NO** $\rightarrow$ contradiction!
5. **If contradiction**: Flip that diode’s state, re-solve from Step 2.

The Door Analogy for Assume-and-Verify

Imagine you’re trying to figure out if doors in a hallway are open or closed, but you can’t see them.

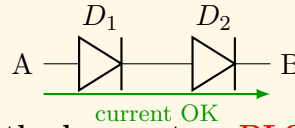
1. **Guess**: “I think door #1 is open and door #2 is closed.”
2. **Check**: Walk through the hallway with that assumption. If you find wind pushing door #1 in the wrong direction (contradiction), your guess was wrong.
3. **Fix**: Change that door’s state and try again.

With N diodes, there are 2^N possible states, but contradictions eliminate wrong ones quickly. For 1 diode: at most 2 tries. For 2 diodes: at most 4 tries.

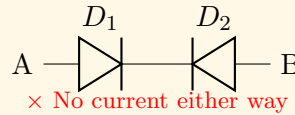
2.4 Series Diode Combinations (Fig. 3.4)

Three Configurations — Which Conducts?

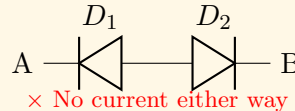
(a) Both face right — **CONDUCTS**



(b) Cathodes meet — **BLOCKED**



(c) Anodes meet — **BLOCKED**



Why (b) and (c) fail — Physical Intuition

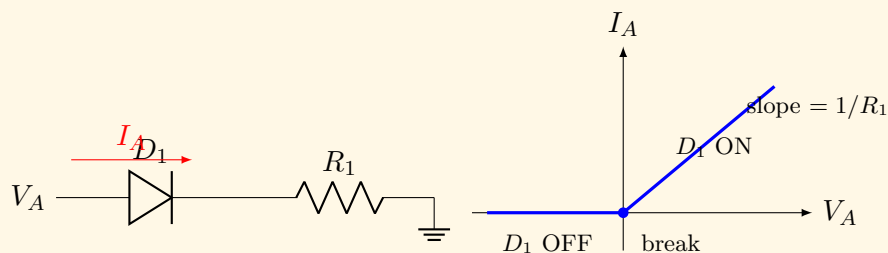
(b) Cathodes meeting: Try current $A \rightarrow B$: D_1 is happy (anode $\rightarrow$ cathode) but then current must enter D_2 at its cathode — **blocked**. Try $B \rightarrow A$: D_2 is happy but D_1 blocks. *Two one-way doors facing each other = deadlock.*

(c) Anodes meeting: Same logic, mirrored. Both doors face outward from the meeting point. Neither lets current through.

General rule: N series diodes conduct **only** when **all** point the same direction. That's 2 out of 2^N possible arrangements.

2.5 Diode-Resistor I/V Characteristic (Fig. 3.7)

Series D_1 - R_1 Circuit: The Building Block of All Rectifiers



Detailed Assume-and-Verify Analysis:

Case 1: $V_A < 0$

1. Assume D_1 OFF (open circuit)
2. Circuit: just V_A connected to nothing $\Rightarrow I_A = 0$
3. Verify: $V_D = V_A - 0 = V_A < 0$ (reverse biased) ✓

Case 2: $V_A > 0$

1. First try: Assume D_1 OFF $\Rightarrow I_A = 0$, no drop across R_1 , so $V_D = V_A > 0$
2. **CONTRADICTION:** Diode has $V_D > 0$ but we assumed OFF!
3. Re-assume D_1 ON (short): $I_A = V_A/R_1 > 0$ ✓ (current flows anode $\rightarrow$ cathode)

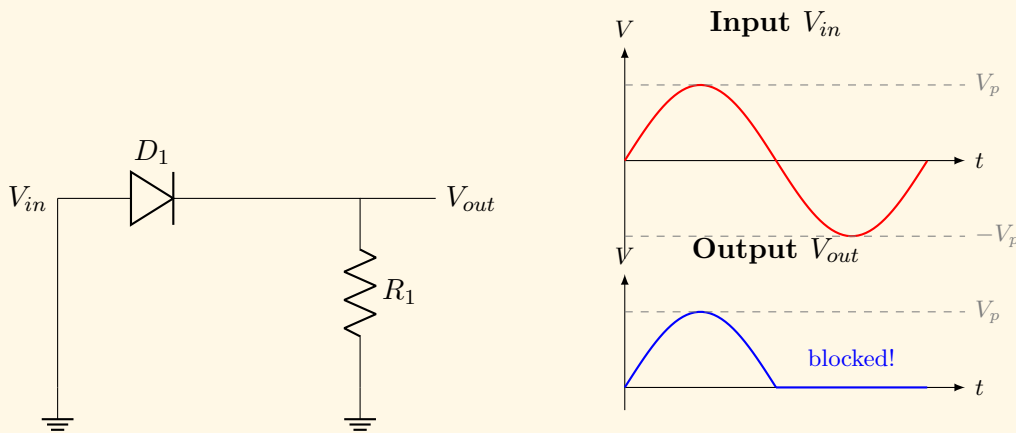
$$I_A = \begin{cases} 0 & V_A < 0 \\ V_A/R_1 & V_A \geq 0 \end{cases}$$

Why This Matters

This simple diode+resistor is the **fundamental building block** of every rectifier. The resistor limits the current (preventing infinite I_D), and the diode selects which polarity passes through. Every circuit in Section 3.5 is built from this idea.

2.6 Half-Wave Rectifier (Fig. 3.10) — CORE APPLICATION

Half-Wave Rectifier: Circuit, Input, and Output Waveforms



Step-by-step operation:

1. **Positive half-cycle** ($V_{in} > 0$): Anode of D_1 is at $V_{in} > 0$, cathode connects to R_1 (near ground). So $V_D > 0 \Rightarrow D_1$ ON (short). Output follows input: $V_{out} = V_{in}$.
2. **Negative half-cycle** ($V_{in} < 0$): Anode at negative voltage, cathode near ground. $V_D < 0 \Rightarrow D_1$ OFF (open). No current flows, no drop across R_1 : $V_{out} = 0$.

$$V_{out} = \begin{cases} V_{in} & V_{in} \geq 0 \\ 0 & V_{in} < 0 \end{cases}$$

The Turnstile Analogy for Rectification

Imagine people (current) trying to pass through a turnstile (diode) in both directions. During the positive half-cycle, people push forward — the turnstile lets them through. During the negative half-cycle, people try to push backward — the turnstile locks. Result: people only flow in one direction, creating a net positive flow (DC component).

2.6.1 Average Voltage Derivation — Step by Step

Full Mathematical Derivation of $V_{out,avg}$

The average of any periodic signal over one period T :

$$V_{avg} = \frac{1}{T} \int_0^T V_{out}(t) dt$$

For our half-wave rectifier, $V_{out} = V_p \sin \omega t$ for $0 \leq t \leq T/2$ and $V_{out} = 0$ for $T/2 \leq t \leq T$:

$$V_{out,avg} = \frac{1}{T} \int_0^{T/2} V_p \sin \omega t dt + \frac{1}{T} \underbrace{\int_{T/2}^T 0 dt}_{=0} \quad (3.5)$$

$$= \frac{1}{T} \cdot \frac{V_p}{\omega} \left[-\cos \omega t \right]_0^{T/2} \quad (3.7)$$

$$= \frac{1}{T} \cdot \frac{V_p}{\omega} \left(\underbrace{-\cos(\omega \cdot T/2)}_{-\cos \pi = +1} + \underbrace{\cos(0)}_{=1} \right)$$

$$= \frac{1}{T} \cdot \frac{V_p}{\omega} \cdot 2 = \frac{2V_p}{T\omega}$$

$$= \frac{2V_p}{(2\pi/\omega) \cdot \omega} = \boxed{\frac{V_p}{\pi} \approx 0.318 V_p} \quad (3.8)$$

Intuition: Why V_p/π ?

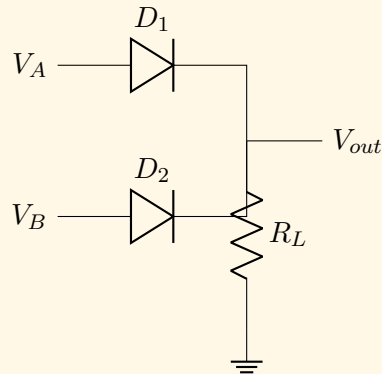
A full half-sine wave (just the bump) has an average of $2V_p/\pi \approx 0.637 V_p$ over its own half-period. But we're averaging over the *full* period which includes the zero second half. So:

$$V_{avg} = \frac{1}{2} \cdot \frac{2V_p}{\pi} = \frac{V_p}{\pi}$$

Numerical example: If $V_p = 5 \text{ V}$, then $V_{avg} = 5/\pi \approx 1.59 \text{ V}$. Not great for powering a 3.3 V device — we need smoothing (Section 3.5)!

2.7 The Diode OR Gate (Fig. 3.8)

Diode Logic: OR Gate



Case: $V_A = 3 \text{ V}$, $V_B = 0 \text{ V}$:

D_1 : anode at 3 V. If ON, $V_{out} = 3$ V. Then D_2 : anode at 0 V, cathode at 3 V $\Rightarrow V_{D2} = -3$ V $\Rightarrow$ OFF ✓.

Result: $V_{out} = \max(V_A, V_B)$ — this is a logical **OR gate**!

The higher input “wins” because its diode forward-biases first, pulling V_{out} high and reverse-biasing the other diode.

3 Section 3.1.3: Limiters (Clippers)

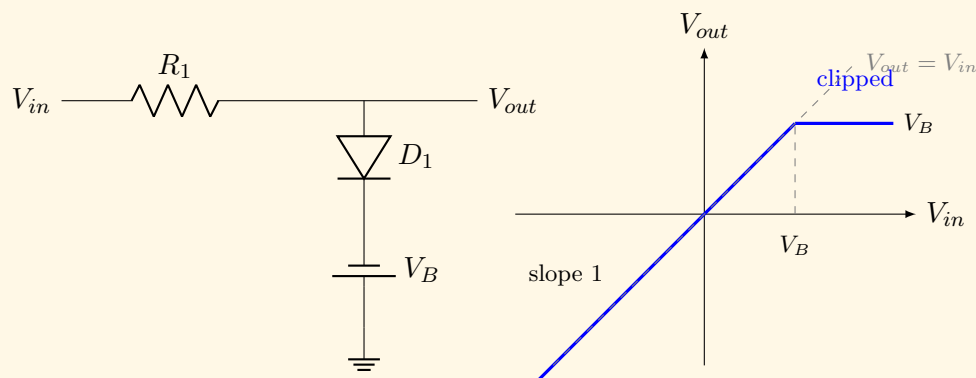
What is a Limiter?

A limiter (or clipper) is a circuit that **clips** the output voltage at a predetermined level. When the input tries to go beyond that level, a diode turns on and “absorbs” the excess, preventing the output from rising further.

Real-world use: Protecting sensitive circuits from voltage spikes. Your phone’s input has limiters to prevent damage from static discharge.

3.1 The Diode-Battery Combination (Fig. 3.11)

Positive Limiter at $+V_B$ — Circuit and Transfer Characteristic



How it works (step by step):

1. $V_{in} < V_B$: The output is less than V_B , so D_1 's anode (V_{out}) < cathode (V_B) $\Rightarrow D_1$ OFF. No current through D_1 , no current through R_1 either, so $V_{out} = V_{in}$.
2. V_{in} **approaches** V_B : V_{out} reaches V_B and D_1 's anode equals cathode.
3. $V_{in} > V_B$: D_1 turns ON (short). Now V_{out} is clamped to V_B regardless of how high V_{in} goes. The “excess” current flows through D_1 into the battery.

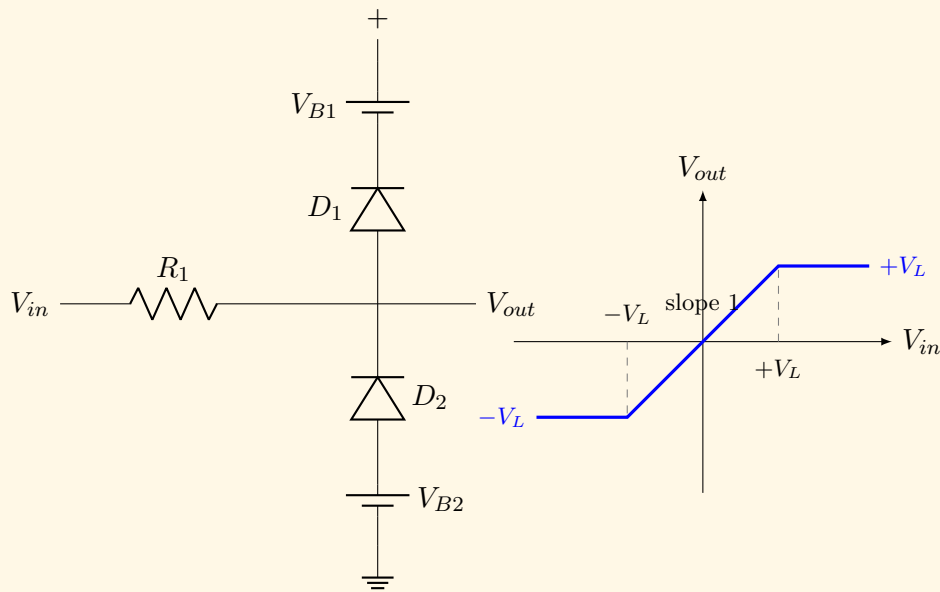
$$V_{out} = \begin{cases} V_{in} & V_{in} < V_B \\ V_B & V_{in} \geq V_B \end{cases}$$

The Pressure Relief Valve

A limiter is like a pressure relief valve on a boiler. Pressure (voltage) builds normally until it reaches the set point (V_B). Once exceeded, the valve opens and vents excess steam (current through D_1), keeping pressure (output voltage) constant.

3.2 Symmetric Limiter (Fig. 3.51)

General Symmetric Limiter at $\pm V_L$ — Clips Both Polarities



$$V_{out} = \begin{cases} -(V_{D,on} + V_{B2}) & V_{in} \leq -(V_{D,on} + V_{B2}) \\ V_{in} & -(V_{D,on} + V_{B2}) < V_{in} < V_{D,on} + V_{B1} \\ V_{D,on} + V_{B1} & V_{in} \geq V_{D,on} + V_{B1} \end{cases}$$

Design rule: To clip at $\pm V_L$: set $V_B = V_L - V_{D,on}$.

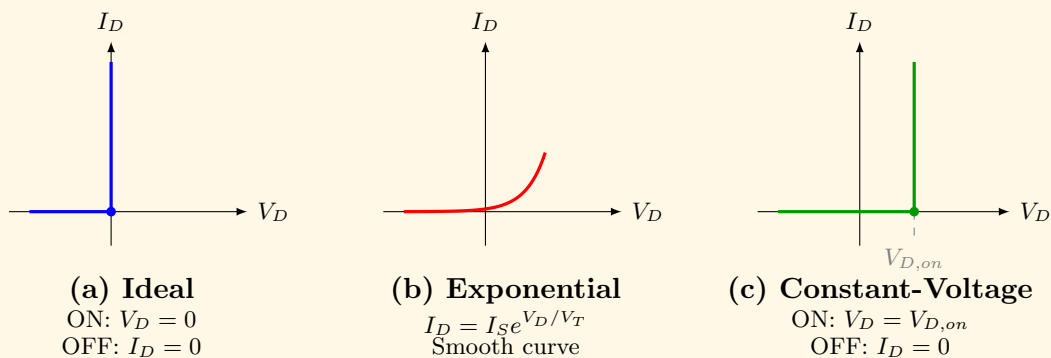
Special case: If $V_L < V_{D,on}$ (e.g., clip at 100 mV with $V_{D,on} = 800$ mV), use **negative** $V_B = V_L - V_{D,on} < 0$. The negative battery opposes the diode drop.

4 Section 3.2: The Constant-Voltage (CV) Model

Why Do We Need a Better Model?

The ideal diode says $V_D = 0$ when ON. But real silicon diodes need ~ 0.7 – 0.8 V to turn on. For circuits with small signals (e.g., 1–2 V), ignoring this 0.7 V drop introduces **significant error**. The CV model adds this reality with minimal complexity.

Three Diode Models Compared — I/V Curves (Fig. 3.13)



When to Use Each Model

Model	Use When	Accuracy
Ideal	Quick intuition, $V_{signal} \gg V_{D,on}$	Low
Constant-Voltage	Most practical problems (DEFAULT)	Good
Exponential	Need exact currents, different diode areas	High (complex)
Small-Signal	AC perturbation around a DC bias point	High for AC

Rule of thumb: If the problem doesn't specify, use **Constant-Voltage** with $V_{D,on} = 0.7\text{--}0.8\text{ V}$.

Key Rules at the ON/OFF Boundary

At the exact switching edge (diode is just about to turn on or off):

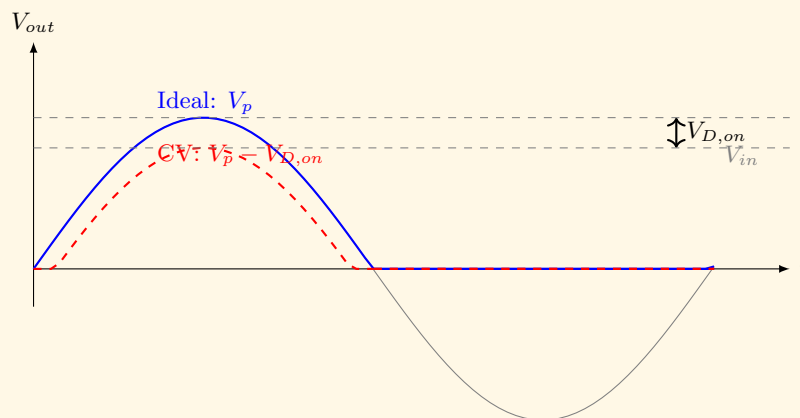
- $I_D = 0$ **AND** $V_D = V_{D,on}$ simultaneously
- Think of it as the instant the valve is just barely touching the stopper

Verification rules (CV model):

- Assumed ON $\rightarrow$ check: $I_D > 0$ (anode $\rightarrow$ cathode)? If $I_D < 0 \Rightarrow$ flip!
- Assumed OFF $\rightarrow$ check: $V_D < V_{D,on}$? If $V_D \geq V_{D,on} \Rightarrow$ flip!

4.1 CV Model Effect on the Half-Wave Rectifier

Half-Wave Rectifier: Ideal vs. CV Model Output Waveforms

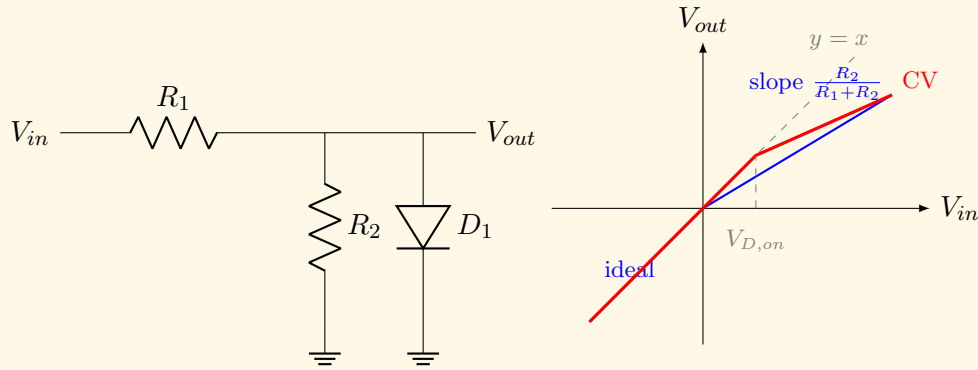


Key differences with CV model:

1. Peak output drops from V_p to $V_p - V_{D,on}$ (lose $0.7\text{--}0.8\text{ V}$)
2. Diode doesn't turn on until $V_{in} > V_{D,on}$ (small dead zone near zero)
3. Average output decreases: $V_{avg} \approx (V_p - V_{D,on})/\pi$

4.2 Voltage Divider with Diode (Fig. 3.14) — Detailed Analysis

Voltage Divider with Diode Clamp — Circuit and Both Models



Region 1 (D_1 OFF): When $V_{out} < V_{D,on}$, diode is open. R_2 carries no relevant current from D_1 . $V_{out} = V_{in}$ (slope 1).

Breakpoint: $V_{in} = V_{D,on}$ (CV model) or $V_{in} = 0$ (ideal).

Region 2 (D_1 ON): D_1 shorts (ideal) or acts as $V_{D,on}$ battery. V_{out} is now a voltage divider:

$$V_{out} = \frac{R_2 V_{in} + R_1 V_{D,on}}{R_1 + R_2} \quad (3.11)$$

Slope after breakpoint: $R_2/(R_1 + R_2) < 1$. The diode “steals” current, reducing the output’s sensitivity to input changes.

5 Section 3.3: The Exponential Model

When the Simple Models Aren’t Enough

The ideal and CV models treat all diodes the same. But what if two diodes have **different sizes** (cross-section areas)? Both models say they’re identical. The exponential model resolves this:

$$I_D = I_S \exp\left(\frac{V_D}{V_T}\right) \quad (2.109)$$

where $I_S \propto$ diode area and $V_T = kT/q \approx 26$ mV at room temperature.

The Highway Analogy for I_S

I_S is like the number of lanes on a highway. A 4-lane highway (large I_S) carries more cars (current) at the same speed limit (voltage) than a 2-lane highway (small I_S). When two highways merge, the wider one carries proportionally more traffic.

5.1 Parallel Diodes with Different Areas (Fig. 3.15)

For $D_1 \parallel D_2$ sharing the same voltage V_D :

Both diodes have $I_D = I_S e^{V_D/V_T}$, so:

$$\frac{I_{D1}}{I_{S1}} = e^{V_D/V_T} = \frac{I_{D2}}{I_{S2}}$$

$$\boxed{\frac{I_{D1}}{I_{S1}} = \frac{I_{D2}}{I_{S2}}} \quad (3.14)$$

Current divides proportionally to I_S :

$$I_{D1} = \frac{I_{in}}{1 + I_{S2}/I_{S1}}, \quad I_{D2} = \frac{I_{in}}{1 + I_{S1}/I_{S2}} \quad (3.15-3.16)$$

Physical Intuition

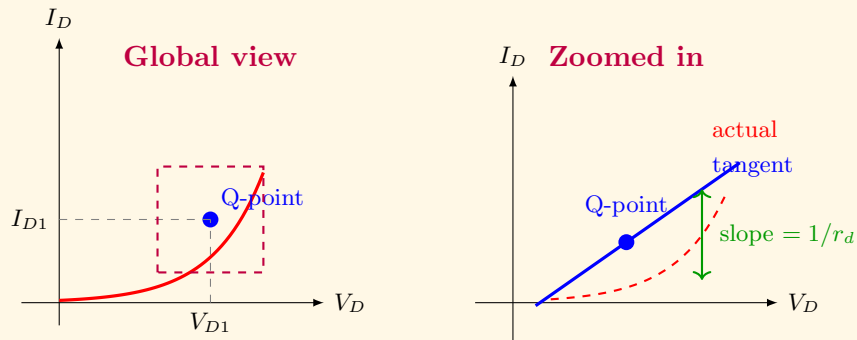
Two parallel diodes at the same V_D are like two pipes of different diameters connected to the same pressure source. The wider pipe (larger I_S) carries more flow. The ratio of flows equals the ratio of pipe areas (I_S values).

6 Section 3.4: Small-Signal Analysis

The Core Idea — Linearization

The diode equation $I_D = I_S e^{V_D/V_T}$ is **nonlinear** (exponential). This makes circuit analysis hard. But if we apply a **small perturbation** $\Delta V_D \ll V_T$ around a DC bias point, the exponential locally looks like a **straight line**. We can then replace the diode with a simple **resistor** r_d .

Graphical View: Zooming Into the Exponential Curve



When you zoom in close enough, the exponential curve looks like a straight line. The slope of that line is $1/r_d = I_D/V_T$, so $r_d = V_T/I_D$.

6.1 Derivation of r_d — Step by Step

If V_D changes from V_{D1} to $V_{D1} + \Delta V$:

$$I_{D2} = I_S \exp \frac{V_{D1} + \Delta V}{V_T} = I_{D1} \exp \frac{\Delta V}{V_T} \quad (3.46)$$

$$\approx I_{D1} \left(1 + \frac{\Delta V}{V_T} \right) \quad \text{for } \Delta V \ll V_T \text{ (Taylor expansion)} \quad (3.48)$$

$$\Rightarrow \Delta I_D = I_{D2} - I_{D1} = \frac{I_{D1}}{V_T} \Delta V \quad (3.49)$$

$$r_d = \frac{\Delta V}{\Delta I_D} = \frac{V_T}{I_{D1}} = \boxed{\frac{26 \text{ mV}}{I_D}} \quad (3.58)$$

Why r_d is Inversely Proportional to I_D

At **higher** currents, you're operating on a **steeper** part of the exponential curve. Steeper slope $\Rightarrow$ more current change per volt $\Rightarrow$ **lower** resistance.

Numerical examples:

I_D	r_d	Interpretation
0.1 mA	260 Ω	Low current: diode acts like a moderate resistor
1 mA	26 Ω	Medium current: fairly low resistance
10 mA	2.6 Ω	High current: almost a short circuit

6.2 The Two-Step Analysis Method (Bias + Signal)

How to Solve Small-Signal Problems

Step 1: DC Bias Analysis (use CV model)

1. Replace diodes with $V_{D,on}$ batteries
2. Find DC operating point: I_D , $V_{out,DC}$
3. Calculate $r_d = V_T/I_D = 26 \text{ mV}/I_D$

Step 2: AC Signal Analysis (use r_d)

1. Turn off all DC sources (voltage sources $\rightarrow$ short, current sources $\rightarrow$ open)
2. Replace each forward-biased diode with r_d
3. Solve the resulting linear circuit for the AC output

Final answer: $V_{out}(t) = V_{out,DC} + v_{out,AC}(t)$

7 Section 3.5: Applications of Diodes

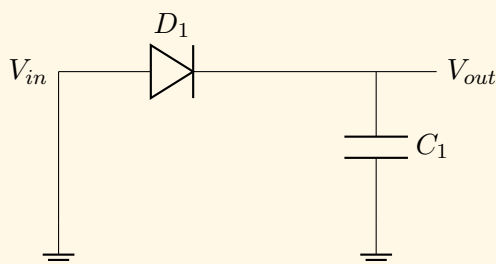
THIS IS THE MOST IMPORTANT SECTION FOR YOUR QUIZ

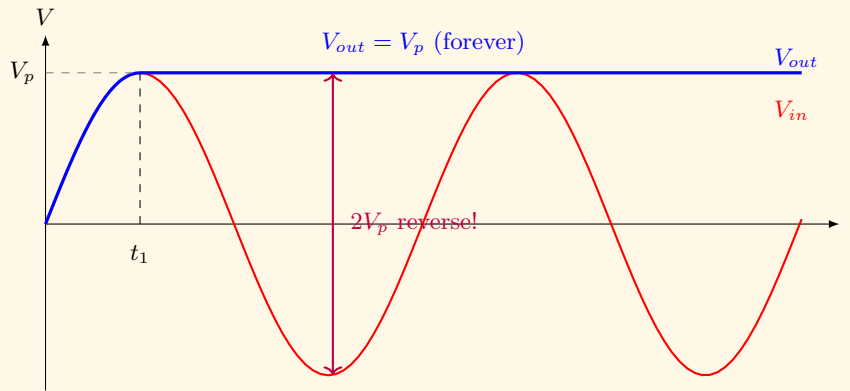
Everything before this was building tools. Now we **use** them. Master these five applications:

1. Half-Wave Rectifier with capacitor smoothing (ripple)
2. Full-Wave Bridge Rectifier
3. Zener Voltage Regulators (line & load regulation)
4. Limiting Circuits with batteries
5. Voltage Doublers and Multipliers

7.1 Section 3.5.1: Peak Detector (No Load) — Fig. 3.30

Peak Detector Circuit and Step-by-Step Waveform





Timeline (using CV model):

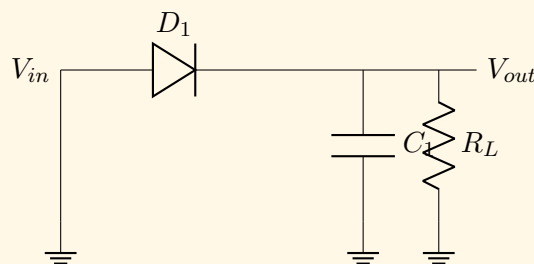
1. $t < t_1$: V_{in} rises. When $V_{in} > V_{D,on}$, D_1 turns ON. Current flows into C_1 , charging it. V_{out} follows $V_{in} - V_{D,on}$.
2. $t = t_1$ (**peak**): $V_{in} = V_p$, $V_{out} = V_p - V_{D,on}$.
3. $t > t_1$: V_{in} starts falling. If V_{out} tried to fall, C_1 would need to discharge through D_1 's cathode $\rightarrow$ **impossible** (blocked!). So D_1 turns OFF.
4. $t = t_3$: $V_{in} = -V_p$. Reverse voltage across D_1 : $V_{D1} = V_{in} - V_{out} = -V_p - (V_p - V_{D,on}) \approx -2V_p$.
5. **Forever after**: V_{out} stays at $V_p - V_{D,on}$ (no discharge path).

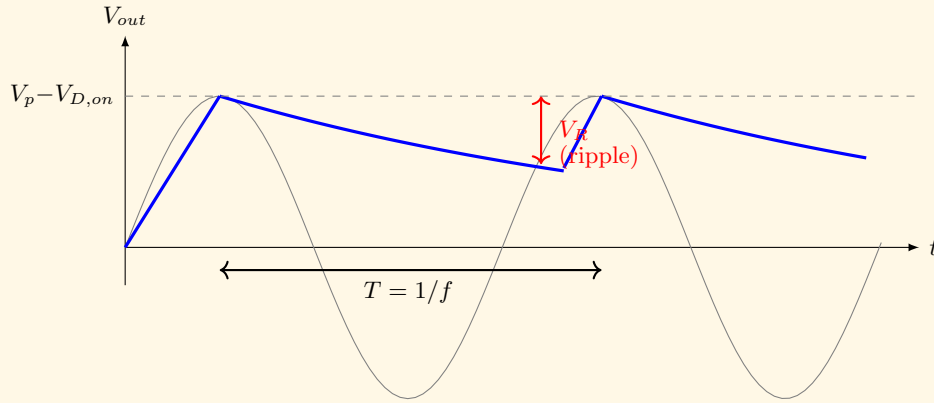
Critical Design Constraint

Diodes in peak detector circuits must have a **breakdown voltage** $> 2V_p$! If the diode breaks down at $-2V_p$, current will flow backward and the circuit fails.

7.2 Section 3.5.1: Rectifier with Load — Ripple

Half-Wave Rectifier with C_1 and R_L — The Ripple Problem





The Leaky Bucket Analogy for Ripple

Imagine filling a bucket (capacitor) from a periodic fountain (rectified AC). Between refills, the bucket leaks through a hole (load R_L):

- **Big bucket, small hole:** Water level barely drops between refills $\Rightarrow$ small ripple (large C_1 , large R_L)
- **Small bucket, big hole:** Water level drops significantly $\Rightarrow$ large ripple (small C_1 , small R_L)

7.2.1 Ripple Derivation — Full Steps

Detailed Derivation

1. **Discharge equation:** After peak (t_1), D_1 is OFF. C_1 discharges through R_L :

$$V_{out}(t) = (V_p - V_{D,on}) \exp\left(\frac{-t}{R_L C_1}\right) \quad (3.75)$$

2. **Linearization:** If $R_L C_1 \gg T$ (capacitor is “large”), the discharge is small:

$$\exp(-\varepsilon) \approx 1 - \varepsilon \quad \text{for } \varepsilon \ll 1$$

$$V_{out}(t) \approx (V_p - V_{D,on}) \left(1 - \frac{t}{R_L C_1}\right) \quad (3.76)$$

3. **Ripple = drop over one period T :** The capacitor discharges for nearly the full period $T = 1/f$:

$$V_R = (V_p - V_{D,on}) \cdot \frac{T}{R_L C_1} = \boxed{\frac{V_p - V_{D,on}}{R_L C_1 f}} \quad (3.80)$$

4. **Alternative form** (using load current I_L):

$$I_L = \frac{V_p - V_{D,on}}{R_L} \Rightarrow V_R = \frac{I_L}{C_1 f} \quad (3.83)$$

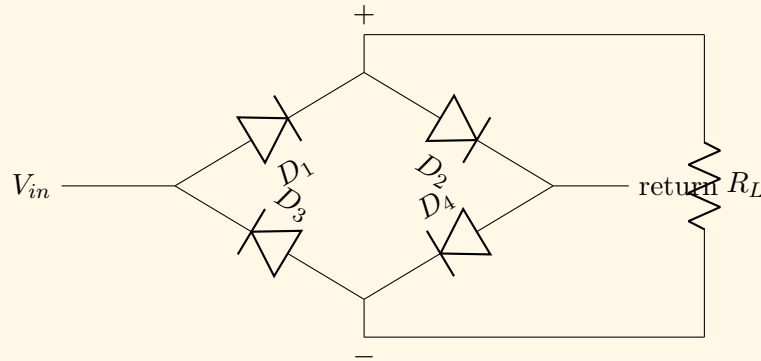
Why Does Ripple Depend on These Quantities?

- **Larger $C_1 \Rightarrow$ smaller ripple:** More stored charge = slower discharge (bigger bucket)
- **Larger $R_L \Rightarrow$ smaller ripple:** Less load current = slower discharge (smaller leak)
- **Higher $f \Rightarrow$ smaller ripple:** Less time between refills (fountain cycles faster)

- **Larger $V_p \Rightarrow$ larger ripple:** Higher voltage = more current = faster discharge

7.3 Section 3.5.1: Full-Wave Bridge Rectifier (Fig. 3.38)

Full-Wave Bridge Rectifier and Current Paths



Positive half ($V_{in} > 0$): D_3 and D_4 ON, D_1 and D_2 OFF.

Path: $V_{in}(+) \rightarrow D_3 \rightarrow R_L \rightarrow D_4 \rightarrow V_{in}(-)$. Output: $V_{out} = V_{in} - 2V_{D,on}$.

Negative half ($V_{in} < 0$): D_1 and D_2 ON, D_3 and D_4 OFF.

Path: $V_{in}(-) \rightarrow D_1 \rightarrow R_L \rightarrow D_2 \rightarrow V_{in}(+)$. Output: $V_{out} = -V_{in} - 2V_{D,on}$.

Result: Both halves produce positive output $\Rightarrow$ effective ripple frequency doubles!

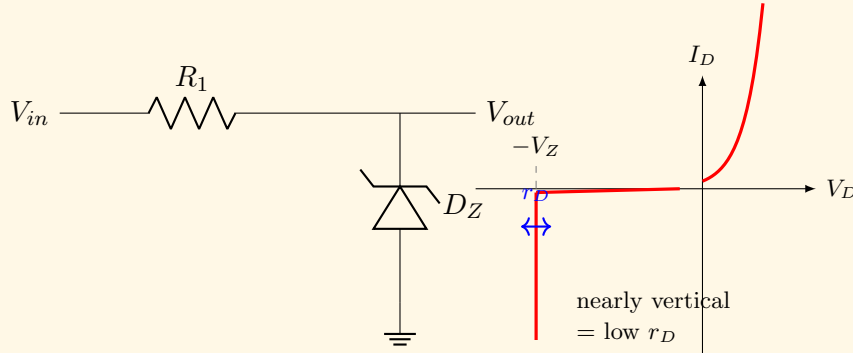
Half-Wave vs. Full-Wave — Complete Comparison

	Half-Wave	Full-Wave Bridge
Diodes	1	4
$V_{D,on}$ lost per cycle	$V_{D,on}$	$2V_{D,on}$
Ripple	$\frac{V_p - V_{D,on}}{R_L C_1 f}$	$\frac{V_p - 2V_{D,on}}{2R_L C_1 f}$ (HALF!)
Max reverse voltage	$2V_p$	$\sim V_p$ (HALF!)
Avg output (no cap)	V_p/π	$2V_p/\pi$
Practical use	Simple, cheap	Most power supplies

Why is FW ripple halved? The capacitor discharges for only **half** a period before the next peak arrives (since both half-cycles contribute).

7.4 Section 3.5.2: Zener Voltage Regulators

Zener Regulator Circuit and I/V Characteristic



The Dam Spillway Analogy for Zener

A Zener regulator is like a dam with a spillway set at a specific height:

- **Water level < spillway height:** All water stays behind dam (no regulation, Zener not in breakdown)
- **Water level = spillway height:** Excess water overflows via spillway. No matter how much rain falls upstream (V_{in} changes), the water level behind the dam (V_{out}) stays constant.
- The spillway has a tiny slope (r_D) — the water level rises imperceptibly as flow increases.

7.4.1 Line Regulation — How Well Does It Regulate?

Line Regulation Derivation

Question: If V_{in} changes by ΔV_{in} , how much does V_{out} change?

In the small-signal model, the Zener becomes a resistor r_D :

$$v_{out} = v_{in} \cdot \frac{r_D}{r_D + R_1} \quad (3.100)$$

Example: $R_1 = 1 \text{ k}\Omega$, $r_D = 5 \Omega$:

$$\frac{v_{out}}{v_{in}} = \frac{5}{1005} = 0.005 = 0.5\%$$

A 1 V input change causes only 5 mV output change — **excellent** regulation!

7.4.2 Load Regulation

$$\left| \frac{v_{out}}{i_L} \right| = (r_{D1} + r_{D2}) \parallel R_1 \quad (3.108)$$

Interpretation: How much output voltage drops per unit of load current drawn. Lower is better.

7.5 Section 3.5.4: Voltage Doublers and Multipliers

7.5.1 Capacitor Coupling Rules

Two Rules You MUST Know

Rule 1 — Floating Capacitor: If one plate's voltage changes by ΔV , the other plate changes by the **same** ΔV .

Why? $Q = C \cdot V_{across}$. If charge doesn't change (no current path), and one plate shifts, the voltage across C stays constant, so the other plate must shift equally.

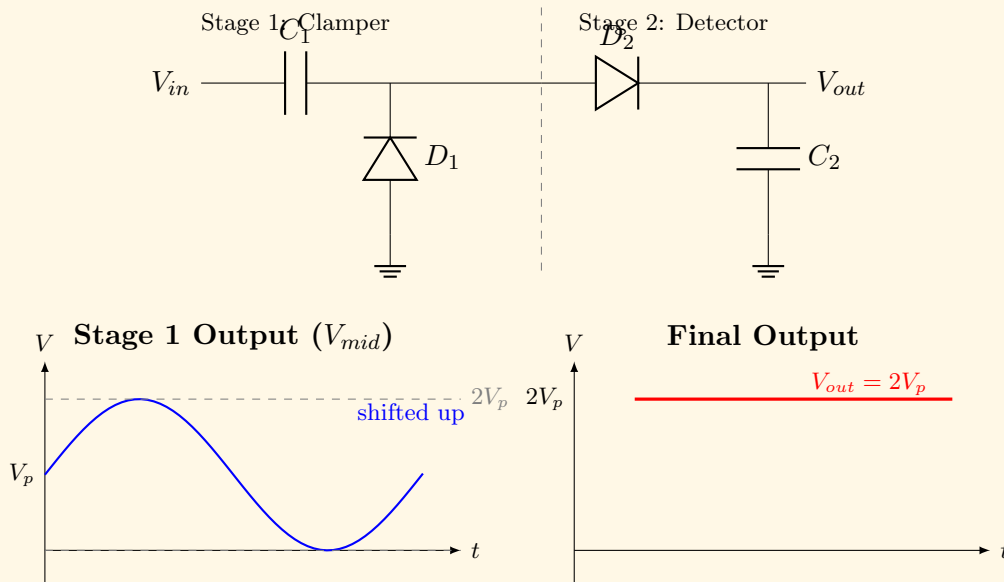
Rule 2 — Capacitive Divider:

$$\Delta V_{out} = \frac{C_1}{C_1 + C_2} \Delta V_{in} \quad (3.111)$$

CAUTION: C_1 is in the **numerator** — opposite of resistive dividers! Small $C_2 \Rightarrow$ large output (charge conservation, not Ohm's law).

7.5.2 Voltage Doubler — Two Stages

Voltage Doubler: Clamper + Peak Detector



Step-by-step:

1. **Stage 1 (Clamper):** During negative input swing, D_1 ON, charges C_1 to V_p . After charging, $V_{mid} = V_{in} + V_p$. This signal swings from 0 to $2V_p$.
2. **Stage 2 (Peak Detector):** $D_2 + C_2$ capture the $2V_p$ peak.

$$V_{out} = 2V_p - 2V_{D,on}$$

Voltage Tripler / N-tupler

Stack more clamper+detector stages:

- **Tripler:** 3 stages $\Rightarrow V_{out} = 3V_p - 3V_{D,on}$
- **N-tupler:** N stages $\Rightarrow V_{out} = NV_p - NV_{D,on}$

Each stage adds V_p DC shift but loses one $V_{D,on}$. Diminishing returns for large N .

7.6 Level Shifters

A forward-biased diode maintains $\approx V_{D,on}$ across it. This acts as a constant DC shift:

- 1 diode: shifts by $V_{D,on} \approx 0.8\text{ V}$
- N diodes in series: shifts by $N \cdot V_{D,on}$

Real-world use: In multi-stage amplifiers, DC levels between stages may not match. Diode strings provide clean, predictable voltage offsets with low AC impedance (r_d).